

# MEMORY IS SIGNED: CAUSAL CREDIT MEASUREMENT FOR HISTORICAL REPRESENTATIONS IN VISION-LANGUAGE GUI AGENTS

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## ABSTRACT

A GUI agent that keeps its history can still act on a record it has already superseded, with the correcting evidence sitting in the same context window. We show that this failure is a property of the model’s *historical key-value representations*, and that its direction is measurable and predictable. Replacing one historical key span in a frozen vision-language model, while holding screenshots, action candidates and recipient positions fixed, moves critical-decision accuracy from 60% to 100% when the replaced span is a superseded update, and from 60% to 0% when it is the latest one. Across 200 controlled histories and 1,000 matched replacements, superseded updates carry negative credit in 92% of cases and latest updates carry positive credit in 98.5%, while matched reference spans are indistinguishable from zero; 181 of 200 prefixes exhibit both signs at once. The sign is set by a block’s relation to the current decision, not by its age: an initial record is just as old as the superseded update it precedes, yet its credit is weakly positive. We then show that this quantity is invisible to ordinary training signals. On 400 paired MiniWoB cases, return-weighted training raises the candidate margin the objective optimises from 0.1502 to 0.1711 while completing no more tasks than the frozen base and four fewer than plain action cross-entropy. We introduce SIGMA, a framework that treats measured replacement effects as supervision: a history-aware controller predicts signed credit from clean prefix features and conditions a low-rank correction on historical key-value spans. SIGMA operationalises the measurement; its end-task benefit is not yet established, and we report that boundary explicitly.

## 1 INTRODUCTION

An agent reads an order, observes a correction, and later submits the order using the old value. Nothing was lost. Both the original record and the correction are in the context window, and the newest screenshot is the one in front of the model. What failed was not retention but *influence*: a historical representation that should have stopped affecting the decision did not.

GUI agents are built to accumulate exactly this kind of state. Screenshots, prior actions and past observations are concatenated into a growing prefix, and the resulting key-value (KV) cache is carried forward as the agent’s memory (Wang et al., 2024; Liu et al., 2026). The field’s default assumption is that this memory is monotone: more retained history is more capability, and the way to improve an agent’s use of its past is to improve what it does with the *current* screen. Memory research in this setting has accordingly focused on what to retrieve and how much to trust it (Wang et al., 2024; Zhang et al., 2026). The complementary question—whether a representation that is still present should still be allowed to vote—is rarely asked, because it has no obvious observable. Task success is a single scalar at the end of an episode and attributes nothing to any particular part of the past.

This paper measures that contribution directly. Our starting point is a controlled setting in which the failure is unambiguous, and which we can perturb one representation at a time.

**The problematic observation.** *Interference Chain* presents a frozen vision-language model (VLM) with a sequence of GUI updates: an initial record, a reference observation, an update that a later observation supersedes, a second reference, and a latest update. The final screen offers eight neutrally

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### Same screenshots. Opposite effects.

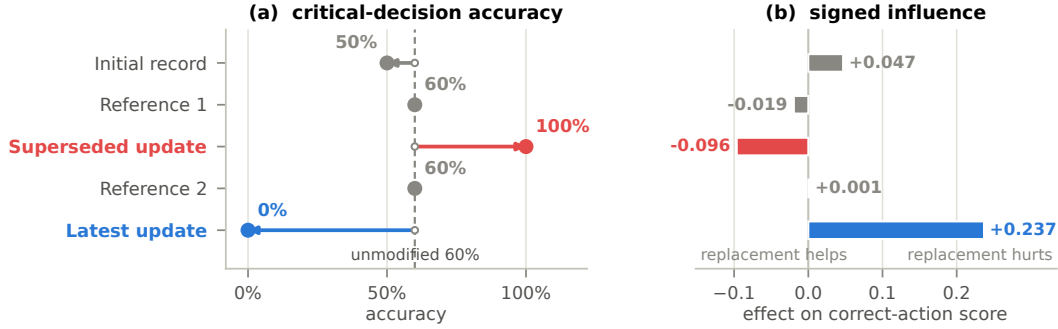


Figure 1: **One historical key span decides the action.** Ten paired prefixes, frozen Qwen2.5-VL-3B-Instruct, decoder key-projection outputs at layers 12–23. Screenshots, candidate slots, token counts and recipient positions are identical across rows; only the source of one historical block’s key span changes. (a) Critical-decision accuracy after replacing that block with a matched donor. (b) The same replacement’s signed effect on the correct-action score of Eq. (3), in the units of Table 1. Replacing the superseded update repairs every failure; replacing the latest update destroys every success; the two matched references leave behaviour unchanged and the older initial record moves it by a single prefix. Age alone therefore does not produce the pattern.

named action slots, and selecting the correct slot requires relating the historical record to the current screen. Screenshots, action candidates, token counts and target positions are held fixed throughout. The only thing we change is which historical key span the decoder attends over: we replace a selected span with a matched one from elsewhere in the same prefix, before rotary position encoding, so the replacement inherits the recipient’s positions.

The result is in Figure 1. The unmodified model selects the correct action in 6 of 10 paired prefixes. Replacing the *superseded* update repairs all four failures, reaching 10 of 10. Replacing the *latest* update destroys all six successes, reaching 0 of 10. Replacing either matched reference leaves accuracy at 6 of 10, and replacing the initial record—an equally old span—moves it only to 5 of 10. Identical pixels, opposite outcomes.

**What this rules out.** The two updates differ in age by one observation, and the initial record is older than both, so recency cannot generate the pattern. Both updates are semantically similar records about the same task, so content alone cannot either. What separates them is their relation to the current decision: one has been superseded and one has not. Age and content are the two signals a memory system can observe without intervening on the model, and neither predicts the sign we measure.

**The measurement does not reduce to confidence.** If interference were merely weak or noisy memory use, better-optimised action scores would fix it. They do not. On 400 paired MiniWoB cases (Figure 5), return-weighted training raises the mean top-1 candidate margin—the quantity such objectives actually optimise—from 0.1502 to 0.1711, the largest margin of any arm. It completes 330 of 400 tasks: one more than the frozen base, and three fewer than plain action cross-entropy, which has the *lowest* margin of all arms at 0.1471. The same decoupling appears in our own precursor runs, where return-conditioned KV adaptation lifted a held-out return margin from 0.1965 to 0.2598 while leaving top-1 action ranking at 0.50 for every arm including the frozen base. An objective can sharpen its own score without changing what the agent does. Reading memory quality off that score is therefore not merely imprecise; it can point the wrong way.

**Contributions.** (i) We introduce a matched-replacement measurement of *signed causal memory credit* for historical KV spans in a frozen GUI VLM, and show the sign is stable and large where the phenomenon predicts and null where it does not (§3). (ii) We show the effect is per-instance, not an artifact of averaging: 181 of 200 prefixes show the superseded and latest signs simultaneously, and each update block is separable from the reference blocks with AUROC 0.86 and 0.94 (§3). (iii) We

show that this signal is not recoverable from the training objectives currently used for GUI memory, using a failure of our own precursor method as evidence (§5). (iv) We give SIGMA, which turns measured credit into supervision for a history-aware controller and a gated low-rank correction (§2).

## 2 MEASURING THE CONTRIBUTION OF A MEMORY

### 2.1 DECISION PREFIXES AND MATCHED MEMORY CREDIT

At GUI step  $t$ , the agent observes  $h_t = (x, o_{1:t}, a_{1:t-1})$ , where  $x$  is the user instruction and  $o_i$  a screenshot. The frozen VLM induces a memory  $M_t = F_\theta(h_t)$ . For selected decoder layers  $\mathcal{L}$ , each historical visual block is indexed by

$$m_{t,j} = \{K_{\ell,j}, V_{\ell,j}\}_{\ell \in \mathcal{L}}, \quad K_{\ell,j}, V_{\ell,j} \in \mathbb{R}^{n_j \times d_\ell}. \quad (1)$$

Heads are flattened into  $d_\ell$  for notation. Current-screen tokens and generated action tokens lie outside the historical intervention mask.

Let  $\mathcal{P}_{j,d}$  replace block  $j$  with a shape-compatible donor  $d$  drawn from the same prefix. For a fixed decision utility  $u$ , the credit of block  $j$  is

$$c_{t,j}^u = u(h_t; M_t) - \mathbb{E}_{d \sim \mathcal{D}_{t,j}} [u(h_t; \mathcal{P}_{j,d}(M_t))]. \quad (2)$$

Positive credit means the original block serves the utility better than its matched replacements; negative credit means the replacements are preferable. The donor distribution, layer set and utility are part of the estimand, and the quantity is conditional: the same screenshot can carry a different sign after another observation or under a different goal. It is not an intrinsic label.

### 2.2 THREE UTILITIES, THREE CLAIMS

The intervention operator is shared but its target depends on available supervision, and we keep three utilities separate throughout.

**Action-score effect.** For a validated target action  $a^*$  of length  $L$ , the diagnostic score is the mean teacher-forced token log-probability

$$u_{\text{act}}(h; M) = \frac{1}{L} \sum_{k=1}^L \log \pi_\theta(a_k^* | h, M, a_{<k}^*). \quad (3)$$

Length normalisation makes this a ranking score, not the log-probability of an executable trajectory. This is the utility used in the ten-prefix pilot.

**Candidate-value effect.** Given a fixed candidate set  $\mathcal{C}(h)$ , scores  $s_a$  from Eq. (3), and independently assigned continuation values  $\bar{Q}(h, a)$ ,

$$p_M^{\mathcal{C}}(a | h) = \frac{\exp(s_a/T_c)}{\sum_{b \in \mathcal{C}(h)} \exp(s_b/T_c)}, \quad u_{\text{val}}(h; M) = \sum_{a \in \mathcal{C}(h)} p_M^{\mathcal{C}}(a | h) \bar{Q}(h, a). \quad (4)$$

The 200-prefix study uses this proxy. Its values describe a redistribution of probability among the specified candidates and depend on candidate coverage and on the continuation policy behind  $\bar{Q}$ . They are *not* on the same scale as Eq. (3), and we never compare the two numerically.

**Trajectory-return effect.** For a restorable environment state  $s_t$ ,

$$u_{\text{roll}}(h_t, s_t; M_t) = \mathbb{E}_{\tau_{t:T} \sim P_\theta(\cdot | h_t, s_t; \text{do}(M_t))} [R(\tau_{t:T})]. \quad (5)$$

This is the performance-linked extension. Neither Eq. (3) nor Eq. (4) substitutes for it when reporting online success, and the estimator’s design is given in Appendix A.

Table 1: **Ten-prefix matched replacement.** Accuracy is the critical-decision accuracy over ten paired prefixes; the effect is the change in the utility of Eq. (3) when that block is replaced. Sign convention: a positive effect means the original block is preferable to its matched replacement. References are patched with donors “Reference 1” and “Reference 2” respectively; update and initial blocks use a reference donor. The two reference rows are the control for the intervention itself.

| Replaced block           | Accuracy    | Change | Signed effect $u_{\text{act}}$ |
|--------------------------|-------------|--------|--------------------------------|
| <i>unmodified model</i>  | 0.60        | —      | —                              |
| Initial record           | 0.50        | −0.10  | +0.0469                        |
| Reference 1              | 0.60        | 0.00   | −0.0194                        |
| <b>Superseded update</b> | <b>1.00</b> | +0.40  | <b>−0.0965</b>                 |
| Reference 2              | 0.60        | 0.00   | +0.0013                        |
| <b>Latest update</b>     | <b>0.00</b> | −0.60  | <b>+0.2368</b>                 |

### 2.3 MEASURING CREDIT WITHOUT TARGET LEAKAGE

Teacher extraction starts from the observable prefix alone: it records visual token spans, candidate-independent decision features, and donor tensors. Each target is patched in a separate evaluation, effects are averaged over admissible donors, and a donor is never the target itself. Recipient length and positions are preserved; content and activation norms need not be. For the training extractor, donors are frozen from a clean prefix pass so an earlier intervention cannot alter a later donor. The within-forward diagnostic reported in §3.1 and this clean-donor variant are different protocols and are recorded separately; we do not relabel the first as the second.

Training actions and evaluators may define labels but never controller inputs. All candidates share prefix-only features, the recurrent state excludes target-action tokens, and test-time control uses predicted credit without correct actions or intervention-derived labels.

## 3 SIGNED CREDIT IS LARGE, STABLE, AND ROLE-DETERMINED

### 3.1 THE TEN-PREFIX INTERVENTION

Table 1 and Figure 1 report the pilot in full. We use Qwen2.5-VL-3B-Instruct (Bai et al., 2025) with frozen parameters, intervening on historical visual tokens at the decoder’s key-projection outputs in layers 12–23 (zero-based, of 36). This follows the activation-patching principle of comparing controlled internal substitutions (Zhang and Nanda, 2024).

The diagnostic separates the roles cleanly on this sample: update-block sign accuracy 1.0, localisation of updates against references at AUROC 1.0, and reference neutrality accuracy 1.0 at a tolerance of 0.025. With ten paired prefixes these are pilot-level statistics and we do not treat them as estimates of a population. Their function is to establish that the protocol yields the intended contrast at all, which licenses the larger study below.

### 3.2 TWO HUNDRED HISTORIES

The pilot leaves open whether the pattern survives scale. We extract credit for all five blocks over 200 independently generated prefixes in a held-in template, giving 1,000 matched-block measurements. Two reference donors are used for non-reference blocks and one other reference donor for a reference block, so reference neutrality is measured against a weaker control than the update blocks are, and we treat it accordingly.

Table 2 and Figure 2 give the result. The two update blocks carry large, opposite, precisely estimated credit: the superseded update is negative in 92.0% of prefixes and the latest update positive in 98.5%, and both bootstrap intervals exclude zero by a wide margin. The three non-update blocks do not. Reference means are within  $1.4 \times 10^{-4}$  of zero and their intervals straddle it.

The initial record is the informative case. It is older than the superseded update, and a recency account predicts it should be at least as harmful. It is not: its mean credit is weakly positive at +0.000798 with

Table 2: **Signed credit over 200 controlled histories.** The utility is the candidate-value proxy of Eq. (4), in candidate-value units; these numbers are not comparable to the action-score effects of Table 1. The interval is a bootstrap over prefixes ( $B = 4000$ ), which is a cluster resample here because each prefix contributes exactly one measurement per role. “Positive” is the fraction of the 200 per-prefix values above zero.

| Historical block         | Mean credit      | Positive     | Bootstrap 95%          | Mean  credit |
|--------------------------|------------------|--------------|------------------------|--------------|
| Initial record           | +0.000798        | 55.0%        | [+0.000178, +0.001429] | 0.003577     |
| Reference 1              | +0.000103        | 50.0%        | [−0.000452, +0.000666] | 0.003035     |
| <b>Superseded update</b> | <b>−0.007612</b> | <b>8.0%</b>  | [−0.008460, −0.006809] | 0.007834     |
| Reference 2              | −0.000135        | 50.0%        | [−0.000762, +0.000464] | 0.003549     |
| <b>Latest update</b>     | <b>+0.011020</b> | <b>98.5%</b> | [+0.010167, +0.011894] | 0.011047     |

### The sign is set by role, not by age.

200 controlled prefixes · 1,000 matched-block measurements · frozen Qwen2.5-VL-3B, decoder K, layers 12–23

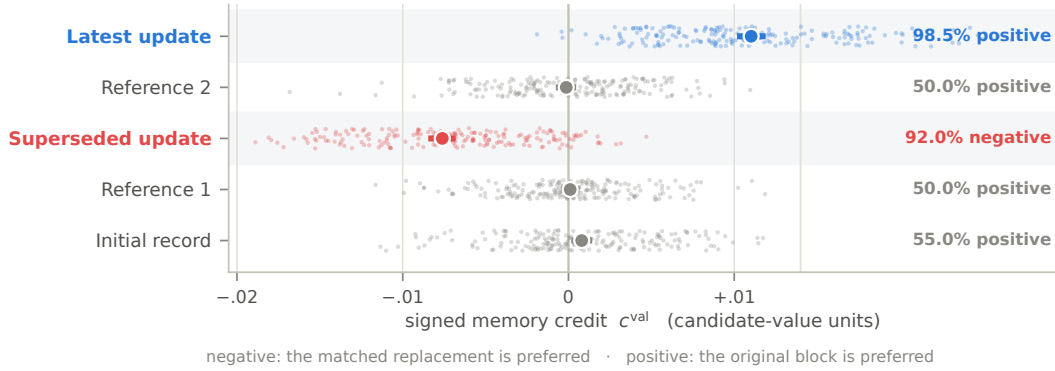


Figure 2: **Signed memory credit by block role over 200 controlled prefixes.** Each faint point is one prefix’s measurement for that block; the bold marker is the mean and the bar a bootstrap 95% interval over prefixes. Blue denotes that the original block is preferable to its matched replacement, red that the replacement is preferable. The two updates carry opposite signs at high precision; the two matched references and the older initial record do not. Values are in candidate-value units (Eq. (4)) and are not on the same scale as Table 1.

an interval excluding zero, and it is positive in 55.0% of prefixes. Whatever marks the superseded update is therefore not its age.

### 3.3 THE EFFECT IS PER-INSTANCE

A near-zero mean can hide large effects of opposing sign, so the aggregate in Table 2 is not by itself evidence that references are inert. We therefore examine the joint pattern per prefix (Figure 3). Of the 200 prefixes, 181 (90.5%) show the superseded update negative *and* the latest update positive simultaneously; the remaining 19 show one of the two signs; none shows neither. The mean absolute effect sizes support the same reading—references sit near 0.0030–0.0035 while the updates sit at 0.0078 and 0.0110—so reference neutrality is not an artifact of cancellation.

Credit also localises each update block against the reference blocks individually: AUROC 0.864 for identifying the superseded update and 0.936 for the latest, using the sign of a single block’s credit with no access to labels. A memory controller that had to decide which block to trust, using only this measurement, would therefore be working from a signal with real discriminative content rather than a role label copied from the data generator.

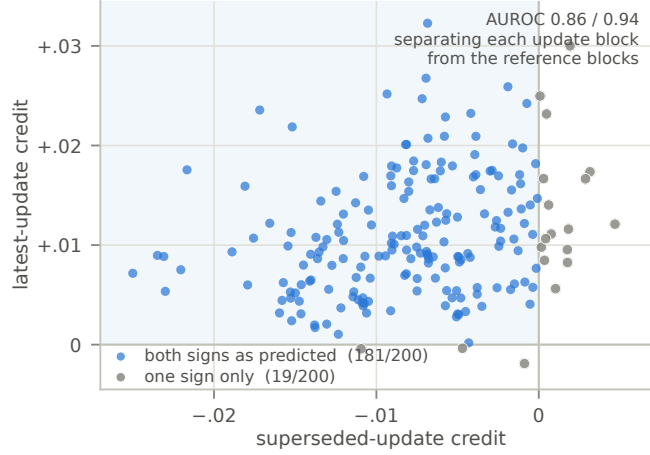


Figure 3: **The opposing pattern holds per prefix.** Each point is one controlled prefix, placed by its superseded-update credit (horizontal) and latest-update credit (vertical). The shaded quadrant is the region a role-driven account predicts. Blue points show both predicted signs, grey show exactly one, and no prefix falls outside both. AUROC values are the discriminability of each update block against the three non-update blocks from credit sign alone.

## 4 SIGMA: TURNING MEASUREMENT INTO ADAPTATION

The measurement solves an identification problem: it says which historical blocks deserve to influence the current decision. It does not say how to change them. SIGMA separates the two, using credit as supervision for a controller and learning the correction direction from the action objective.

### 4.1 HISTORY-AWARE CREDIT DISTILLATION

Memory utility depends on relationships among observations. A code in an old record does not identify itself as superseded; a later update establishes that relation. For each block we pool clean projection features over visual tokens and selected layers, project them to a compact embedding  $z_j$ , and encode order:

$$z_j = P_\phi(\text{Pool}(m_{t,j})), \quad s_j = \text{GRU}_\phi(z_j, s_{j-1}), \quad \hat{c}_{t,j} = f_\phi(z_j, s_j, q_t), \quad (6)$$

where  $q_t$  is a clean prefix-only decision feature and the final recurrent state lets an earlier block’s prediction depend on all observations available at step  $t$ . The GRU is a compact implementation choice, not a claim; it is compared against an independent MLP and a small Transformer.

Targets are scaled by a positive constant  $s_u$  fitted on training labels of the same utility type,  $\tilde{c}^u = c^u / s_u$ ; we never mix action-score and candidate-value units or force the mean reference effect to zero. A neutrality threshold  $\epsilon_u$  defines a three-way target  $y \in \{-, 0, +\}$ . With nonnegative reliability weights  $w_{t,j}$ ,

$$\mathcal{L}_{\text{credit}} = \mathbb{E}_{t,j} w_{t,j} [\text{Huber}(\hat{c}_{t,j}, \tilde{c}_{t,j}^u) + \lambda_s \text{CE}(p_\phi^{\text{sign}}, y_{t,j})] + \lambda_r \mathcal{L}_{\text{rank}}. \quad (7)$$

The ranking term compares clearly separated labels within a prefix; uncertain donor comparisons are masked rather than converted into hard sign labels (Appendix C).

### 4.2 FROM SIGNED CREDIT TO A LEARNED CORRECTION

A signed gate controls a low-rank residual on historical projection outputs. For  $X_{\ell,j} \in \{K_{\ell,j}, V_{\ell,j}\}$  and learned  $B_\ell \in \mathbb{R}^{d_\ell \times r}$ ,  $A_\ell \in \mathbb{R}^{r \times d_\ell}$ ,

$$g_{t,j} = \tanh(\hat{c}_{t,j} / T_g), \quad \Delta X_{\ell,j} = \frac{\alpha}{r} g_{t,j} X_{\ell,j} B_\ell A_\ell, \quad X'_{\ell,j} = X_{\ell,j} + \Delta X_{\ell,j}. \quad (8)$$

$B_\ell A_\ell$  is a general low-rank map, not an orthogonal projector, and a negative gate does not universally reduce a block’s attention contribution: scaling a key can raise or lower its logit depending on the

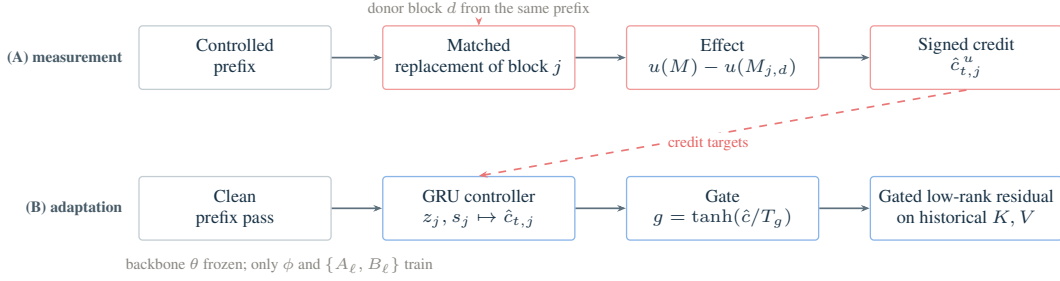


Figure 4: **The SIGMA pipeline.** Lane (A) is the measurement: a frozen model scores a controlled prefix, one historical block’s key span is replaced by a matched donor, and the utility difference averaged over donors becomes the credit target of Eq. (2). Lane (B) is the adaptation that consumes it: a controller reads clean prefix features, predicts credit for every historical block, and turns the prediction into a gate on a low-rank residual applied to historical keys and values of Eq. (8). The two lanes share the intervention operator but not the tensors—the controller never sees donor, effect or target-action information—and the weights of the frozen backbone are never updated.

sign of  $q^\top k$ . The sign of  $c^u$  describes a *finite replacement*, not the derivative of utility along  $XBA$ ; Appendix C gives a two-point counterexample. The credit loss therefore identifies what to adapt, while the action objective learns how:

$$E_{KV} = \frac{1}{|\mathcal{L}|} \sum_{\ell \in \mathcal{L}} \frac{\sum_j \|\Delta X_{\ell,j}\|_F^2}{\sum_j n_j d_\ell}, \quad (9)$$

$$\mathcal{L}_{\text{SIGMA}} = \mathcal{L}_{\text{policy}} + \lambda_c \mathcal{L}_{\text{credit}} + \lambda_E E_{KV}. \quad (10)$$

$A_\ell$  is zero-initialised, so adaptation begins at the base policy. The quadratic term regulates representation change; it is not an optimal-transport objective over source and target distributions.

### 4.3 OPTIMISATION AND DEPLOYMENT

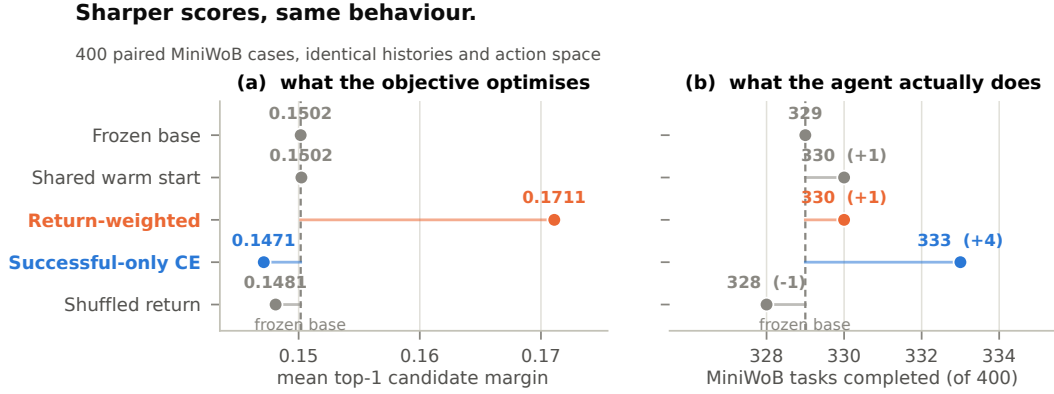
We first fit the controller to measured credit with the residual fixed at zero, then optimise residual and controller jointly with Eq. (10), retaining credit supervision so the gate does not become an unconstrained action adapter. Backbone weights stay frozen. At interaction time a clean prefix pass provides block and decision features, the controller predicts gates once, and a second policy pass applies them before generating an action; new observations produce a new prefix and new gates. This is state-dependent inference, not online weight learning. Both passes are inside the runtime budget, and we do not claim a latency reduction that no implementation has demonstrated. Algorithm 1 states the complete procedure.

## 5 WHY THE OBJECTIVE CANNOT SUBSTITUTE FOR THE MEASUREMENT

If action-score objectives already encoded memory usefulness, Eq. (2) would be an expensive way to recover something training gets for free. We test that directly, using a precursor of our own method as the subject.

**Setup.** SIGMA’s precursor applies return-conditioned KV adaptation to a shared warm start on 41 training trajectories across three AndroidWorld tasks, then trains five variants on identical data: return-weighted, successful-only action cross-entropy, a shared warm start with no further training, and a shuffled-return control. All arms are evaluated on the same 400 paired MiniWoB cases (Rawles et al., 2024).

**Result.** Return-weighted training moves the candidate margin from 0.1502 to 0.1711, the largest of any arm, and ahead of a sign-flipped control at 0.1481 and successful-only CE at 0.1471. It completes 330 of 400 tasks, against 329 for the frozen base and 333 for successful-only CE. Every discrete difference in the entire comparison falls on the click-color family (42, 43, 43, 46, 41 of 100



**Figure 5: The optimised score and the executed behaviour come apart.** Four hundred paired MiniWoB cases, seeds 701–800, four task families, identical histories and action space across arms. (a) The mean top-1 candidate margin. (b) Tasks completed. Return-weighted training produces the largest margin and does not convert it into completions; successful-only cross-entropy produces the smallest margin and the most completions. Panels are dot plots because both quantities have non-zero baselines.

across arms); the other three families are saturated or unchanged. The paired comparison against cross-entropy is 0 rescues and 3 regressions, an exact McNemar  $p = 0.25$  with a family-cluster interval of  $[-2.25, 0]$  percentage points, and all intervals include zero.

**The same decoupling in the precursor itself.** The return-conditioned KV arm lifts a held-out return margin from 0.1965 to 0.2598 over the shared warm start, with sign-flipped labels at 0.1955 and successful-only CE at 0.1874—the direction of the label clearly moves the metric. Top-1 action ranking over those trajectories is 0.50 for the frozen base, the return-conditioned arm, the shuffled control and the cross-entropy arm alike. The metric improved; the ranking did not.

**Interpretation.** We do not conclude that return information is useless for GUI memory. The precursor trains on 41 trajectories with 20 successes, its margin is measured on six held-out trajectories, and Appendix F shows that random labels also improve the same offline score while a first-only history is strongest—so the precursor’s own diagnostic is not trustworthy at this scale. What the two experiments establish is narrower and sufficient for our argument: an objective can sharpen the quantity it optimises while leaving behaviour at or below an untuned baseline, on paired data and within a single seed. Whatever the score is measuring, it is not the agent’s use of its history. That gap is what Eq. (2) is designed to close.

## 6 RELATED WORK

**GUI memory and long-horizon interaction.** Agent Workflow Memory learns reusable action routines and exposes selected workflows to later tasks (Wang et al., 2024). Latent approaches move the interface from text to continuous representations: FocusMem separates content formation, state-conditioned readout and evidence trust under a frozen GUI policy (Zhang et al., 2026), and its trust module explicitly addresses misleading retrieved evidence. We therefore do not claim that selective memory or gating is new. The distinguishing element here is the supervision signal: a measured signed effect of replacing an existing historical decoder span at a fixed decision prefix, rather than a relevance or trust label attached to an external memory item.

**Internal interventions and representation adaptation.** Activation patching measures the consequence of controlled internal substitutions, with conclusions sensitive to the corruption and the metric (Zhang and Nanda, 2024). ReFT learns task-specific representation interventions while base weights stay frozen (Wu et al., 2024), and visual work shows how an internal representation pathology can motivate an architectural remedy (Darcet et al., 2024). SIGMA adopts the same observation-then-

remedy separation; low rank limits the parameter and perturbation budget but is not the source of the causal signal.

**Capability-selective self-distillation.** SPD derives a capability subspace from correctness gradients, steers KV during self-generation, and finetunes on the resulting sequences (Hao et al., 2026). Our target is not a shared generation subspace but the signed contribution of one historical span under the current GUI prefix.

**Memory-focused evaluation.** MemGUI-Bench varies task conditions to test short- and long-term memory use (Liu et al., 2026); the Android-in-the-Wild interaction data underlies this line of device-control evaluation (Rawles et al., 2023), and AndroidWorld evaluates state-changing mobile actions (Rawles et al., 2024); WorkArena++ stresses compositional workflows (Boisvert et al., 2024). These settings motivate measuring task outcomes rather than treating accuracy on a synthetic recognition task as an application result—which is why our own end-task evidence is reported as a boundary rather than a headline.

## 7 DISCUSSION

**What a negative credit means.** It means a specified replacement improves a specified utility under the tested prefix. The superseded record may remain useful for a different question, and a small effect may reflect redundancy rather than irrelevance. Whole-block effects do not decompose additively: removing one supporting block can change another’s contribution. SIGMA predicts credit from the complete observed history for this reason, and its sign is tested under donor and template shifts rather than assumed stable.

**Measurement scope.** Equal-length patching preserves the sequence interface, not the full activation distribution. Donors carry context from elsewhere in the prefix, and effects propagate through later layers. The reported diagnostics are within-forward donors for the pilot and clean-prefill donors for training, recorded separately. Two controls that would tighten the estimand have not been run: a token-position control, and multi-donor sensitivity on held-out templates. We therefore treat the sign and magnitude as established for the donor distributions actually used, and not yet as invariant to them.

**Boundary on the method.** SIGMA’s end-task benefit is not established. On a sealed controlled test of the same family used here, the adapted arm scores 23/200 against 24/200 for its matched control at equal capacity, with a marginally better mean action NLL (0.412004 vs 0.412617); on the native mobile benchmarks, completed episodes stand at 0/21 for the adapted arm against 1/21 for cross-entropy, on a base policy whose failures are dominated by interface and infrastructure faults rather than memory handling. We report SIGMA as a framework that operationalises a validated measurement, not as a benchmark win, and Appendix E gives the full protocol, failure attribution and infrastructure record. The gap between identifying an interfering memory and profitably rewriting it is the central open problem this paper leaves.

## 8 CONCLUSION

GUI memory is not adequately described by what a model retains. In controlled histories, the internal representation of a superseded update interferes with the current decision while the correction sits in the same context window, and the sign of a block’s contribution is set by its relation to the decision rather than by how old it is. That contribution is measurable per instance, is invisible to the objectives currently used to train GUI memory, and is discriminative enough to localise which block to distrust. SIGMA turns it into supervision. The question it leaves open is the one worth asking next: once a memory is identified as interfering, what correction actually helps?

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## AI USE STATEMENT

(This section is required and does not count toward the page limit.)

In this work, we used generative AI tools for the following tasks with required disclosure: developing the conceptual framework and the causal estimand presented in Section 2; formulating the mathematical claims and their notation; proposing and refining the hypotheses tested in Sections 3 and 5; and providing feedback on the design of the analysis methodology. We have not used generative AI tools to generate synthetic datasets, to provide critical ingredients for proving mathematical claims, or to assist in the writing of proofs, and those tasks are not applicable to this work, which contains no formal proofs. Additionally, we used generative AI tools for the following tasks with recommended disclosure: creating and modifying the scientific figures; creating and editing the plotting and evidence-extraction code; drafting and revising parts of the manuscript text; and summarising and analysing existing literature for the related-work discussion. We have reviewed all AI-assisted work. All figures are regenerated from source data by a single extraction script that treats a missing source file as a hard error rather than substituting a default, every empirical quantity in the main text is traced to a named experiment record in Appendix D, and the generated plotting code was executed and its output inspected at final reproduction size. Bibliography entries were carried from the project’s verified citation ledger rather than produced by a language model; one internal inconsistency, a preprint identifier that disagreed with its own URL, was found and corrected against that ledger. We take responsibility for the final content of this work, including text, claims or artifacts produced with the aid of generative AI.

## ETHICS STATEMENT

GUI agents can expose private information or perform consequential actions. Evaluation should use sandboxed benchmark accounts and controlled environments, not users’ production applications. Stored screenshots and trajectories should be minimised, access-controlled and redacted before release. Memory-credit estimation and controller training must not give the agent privileged evaluator state. Deployment beyond benchmarks requires authorisation boundaries and independent checks for irreversible actions.

## REPRODUCIBILITY STATEMENT

Appendix A specifies splits, controller inputs, loss definitions and matched evaluation conditions. Appendix D maps every measured number in the main text to its experiment record. Figures 1, 2, 3 and 5 are generated by code in `plotting/`, which reads the repository JSON records through a single extraction script and prints a verification report; the figure inputs and the table values therefore share one path. The method diagram (Figure 4) is inline vector art in the source. An anonymised snapshot of all of the above accompanies the submission (Appendix G).

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## A TRAINING, INTERVENTION, AND EVALUATION DETAILS

### A.1 A REPRODUCIBLE DEFAULT, NOT A RETROSPECTIVE OPTIMUM

The schedule in Table 3 is the fixed starting protocol for the SIGMA comparison. It is not the configuration that produced the signed-effect measurements of §3, which used frozen Qwen2.5-VL-3B, middle-layer K interventions, and the historical extractor identified in Appendix D.

Table 3: Default protocol and bounded validation search for the learned SIGMA model. No setting here implies a completed run.  $\lambda_E$  applies to the normalised energy of Eq. (9) and cannot be copied numerically from a differently normalised loss.

| Item                            | Specification                                                                       |
|---------------------------------|-------------------------------------------------------------------------------------|
| Primary backbone                | Qwen2.5-VL-3B-Instruct; revision pinned in run manifest                             |
| Scale replication               | Qwen2.5-VL-7B-Instruct, same preprocessing and protocol                             |
| History budget                  | 5 past screenshots + current; 8/12-history sweeps                                   |
| Historical / current pixel caps | 100,352 / 401,408; identical across matched arms                                    |
| Controller                      | 256-dimensional projection and state, one GRU layer, two-layer credit head          |
| Controller alternatives         | Independent MLP and two-layer Transformer, parameter counts reported                |
| Default intervention            | Decoder K, zero-based layers 12–23                                                  |
| Residual                        | Rank 8, $\alpha = 8$ in $\alpha/r$ scaling; zero-initialized output factor          |
| Controller / joint epochs       | 5 / 5; best checkpoint selected on validation task success                          |
| Optimizer                       | AdamW; LR $3 \times 10^{-4}$ (controller), $10^{-4}$ (residual), weight decay .01   |
| Batching                        | 1 prefix, accumulation 8; single model process per GPU                              |
| Training seeds                  | 11, 23, 37                                                                          |
| Evaluation seeds                | 101, 202, 303; reset environment and episode memory each time                       |
| Rank / layer sweep              | $r \in \{4, 8, 16\}$ ; early/middle/late thirds                                     |
| Credit weights                  | $\lambda_c = 1, \lambda_s = .2, \lambda_r = .1$ initially; bounded validation sweep |
| Energy weight                   | $\lambda_E \in \{0, 10, 100, 1000, 3000\}$ for the stated normalization             |
| Gate temperature                | $T_g \in \{.5, 1, 2\}$ after training-only target scaling                           |
| Stage selection                 | No test-based temperature, donor, layer, or epoch selection                         |

### A.2 PREFIX-ONLY FEATURE CONTRACT

For every training prefix, record an immutable identifier, task group, screenshot hashes, image order, prior actions, model revision, prompt, tokenized prefix length, and visual-token spans. Feature extraction stops before target-action tokens. Regression labels, success flags, candidate Q-values, semantic memory roles, and evaluator metadata are not appended to the controller context. All candidate actions at one prefix reuse identical features, and permuting candidate order must not change those features or the predicted credit vector.

### A.3 DONOR CONTROLS AND UNCERTAINTY

Match donors by projection dimension and token-span length; preserve recipient positions and document any interpolation. The principal extractor does not interpolate unequal spans. Use at least two distinct admissible donors where available. Self-patching is an identity control. Same-forward donors and clean-prefill donors are different protocols and have separate output records.

For donor effects  $e_{t,j,d} = u(M_t) - u(\mathcal{P}_{j,d}(M_t))$ , retain every effect, the mean, sign agreement, absolute magnitude, donor identifiers and sampling settings. Bootstrap over donors measures donor sensitivity, not uncertainty over tasks; with two donors it is descriptive. The 200-prefix data use two reference donors for non-reference blocks and one other reference donor for a reference block, so reference-mean neutrality does not establish robust instance-level neutrality—which is why §3 reports per-instance joint signs and absolute magnitudes alongside the means.

#### A.4 NATIVE ROLLOUT CREDIT

For a restorable training state, execute  $K$  paired continuations per intervention arm:

$$\hat{c}_{t,j}^{\text{roll}} = \frac{1}{DK} \sum_{d=1}^D \sum_{k=1}^K [R(\tau^{\text{base},k}) - R(\tau^{\text{patch}(j,d),k})]. \quad (11)$$

Use matched environment state and random-number streams where feasible, fix whether the patch applies only at the current decision or persists into later decisions, and report restoration checks. Real-device retries are not independent samples if state restoration failed. This estimator is specified but has not been executed: every credit measurement in this paper uses the offline utilities of Eqs. (3) and (4), and no native rollout credit on restorable states is reported. The performance-linked version of the estimand therefore remains untested, and §7 treats that as a scope limit rather than a result.

## B TRAINING AND DEPLOYMENT PROCEDURE

Algorithm 1 states the full procedure referenced by §4: measurement of signed credit, controller fitting, joint optimisation of the gated residual, and the deployment loop. The selection rule in line 15 is the one the paper actually uses; no test-based donor, layer or epoch selection is performed.

- 1 **Require:** training prefixes  $\mathcal{H}$ ; frozen VLM  $\theta$ ; donor pool  $\mathcal{D}$ ; utility  $u$
- 2 **Require:** controller  $\phi$ ; residual factors  $\{A_\ell, B_\ell\}_{\ell \in \mathcal{L}}$ ; weights  $\lambda_c, \lambda_E$
- 3 **for** each  $h_t \in \mathcal{H}$  **do**
- 4   Encode the clean prefix; record  $M_t$ , visual spans and decision features  $q_t$
- 5   **for** historical block  $j$  and admissible donor  $d$  **do**
- 6     Evaluate  $u(h_t; M_t) - u(h_t; \mathcal{P}_{j,d}(M_t))$  against the frozen copy of the prefix
- 7   **end for**
- 8   Aggregate effects and donor dispersion into credit targets  $\tilde{c}_{t,j}^u$  and signs  $y_{t,j}$
- 9 **end for**
- 10 Fit  $\phi$  alone on  $\mathcal{L}_{\text{credit}}$  of Eq. (7), with the residual held at identity ( $A_\ell = 0$ )
- 11 **for** each joint epoch **do**
- 12   Sample prefixes; predict  $g_{t,j}$  from clean features only; apply Eq. (8) to historical  $K, V$
- 13   Update  $\phi$  and  $\{A_\ell, B_\ell\}$  on  $\mathcal{L}_{\text{SIGMA}}$  of Eq. (10);  $\theta$  stays frozen
- 14 **end for**
- 15 Select the checkpoint on validation tasks, never on evaluated tasks or donors
- 16 **Deploy:** predict gates once per step from a clean pass; generate; execute; reobserve and rebuild the prefix

Algorithm 1: SIGMA: learning memory adaptation from interventions

## C LOSS DEFINITIONS AND DISTILLATION

**Within-prefix credit ranking.** For pairs whose teacher effects differ by more than a calibrated tolerance, let  $s_{ij} = \text{sign}(\tilde{c}_i - \tilde{c}_j)$  and

$$\mathcal{L}_{\text{rank}} = \frac{1}{|\mathcal{P}|} \sum_{(i,j) \in \mathcal{P}} w_{ij} \log(1 + \exp[-s_{ij}(\hat{c}_i - \hat{c}_j)]), \quad w_{ij} \geq 0. \quad (12)$$

A missing or uncertain label is masked, not silently treated as neutral. Thresholds and target scales are fitted on the training/validation partition for each utility type.

**A finite replacement is not a scaling derivative.** If  $f(x) = x - x^2$ , the original state  $x = 1/2$  has higher utility than the donor  $x = 0$ , yet  $f'(1/2) = 0$  and increasing  $x$  reduces utility. A positive finite replacement credit therefore does not justify an arbitrary positive displacement. In attention, multiplying a key by  $1 + \eta$  changes a query logit by  $\eta q^\top k / \sqrt{d}$ , whose sign depends on  $q^\top k$ . Equation (10) optimises the action effect of the residual separately from the credit predictor for this reason, and the paper nowhere identifies a signed gate with a proven beneficial intervention.

**No-hook policy distillation.** Let  $p_T$  be the adapted teacher and  $p_S$  a same-architecture LoRA student (Hu et al., 2021). Ordinary policy distillation minimises

$$\mathcal{L}_{\text{KD}} = \mathbb{E}_h w_h \text{KL}(\text{sg}[p_T(\cdot | h)] \| p_S(\cdot | h)), \quad w_h \geq 0, \quad (13)$$

with candidate-distribution and token-distribution KL reported as separate interfaces. Residual distillation uses centered logits with  $P = I - \mathbf{1}\mathbf{1}^\top / C$  for  $C$  candidates:

$$\mathcal{L}_{\text{res}} = \mathbb{E}_h w_h \|P(z_S - z_0) - \text{sg}[P(z_T - z_0)]\|_2^2 + \lambda_0 \text{KL}(p_0 \| p_S). \quad (14)$$

Centering removes arbitrary common logit shifts. Signed return baselines are not used as negative coefficients on a KL loss. Static merging is a different operation: a fixed residual  $y' = (I + AB)y$  folds into  $W' = (I + AB)W$ , but a history-dependent factor  $g(h)$  cannot generally be represented by one constant weight matrix, so the earlier merged-student result is not evidence that the recurrent controller has been distilled away. The no-hook student itself has not been trained: it is worth attempting only behind a measured teacher gain, which §7 reports as not established, so we specify the objective and leave it untested rather than claim a deployment saving.

## D EVIDENCE LEDGER

Table 4 maps every measured number in the main text to its record. The extraction script `plotting/extract_evidence.py` reads these files, writes the figure and table inputs, and prints a verification report; a missing source file is a hard error rather than a defaulted value.

Table 4: Evidence boundaries used throughout the manuscript. Record paths are breakable so that no filename can run into the margin.

| Main-text object        | Source record                                      | Scope and limitation                                                 |
|-------------------------|----------------------------------------------------|----------------------------------------------------------------------|
| Fig. 1, Table 1         | tango_v4_interference_chain_kv_credit_pilot10.json | 10 paired prefixes, within-forward matched donors; pilot scale       |
| Table 2, Fig. 2, Fig. 3 | tango_v2_teacher_credit200.json                    | 200 prefixes, 1,000 blocks; candidate-value proxy, not native return |
| Fig. 5                  | tango_miniwob_click4_s701_800.json                 | 400 paired cases; earlier global-return policy, not SIGMA            |
| §5 margin/ranking       | gonogo_current.json                                | 6 held-out trajectories; offline margin only                         |
| §7 boundary             | docs/EXPERIMENT_STATUS_2026-09-17.md               | Sealed controlled test, 3B                                           |
| App. E native status    | docs/SIGMA_BENCHMARK_EXPERIMENT_LOG_2026-09-18.md  | Incomplete episodes; infrastructure faults present                   |

## E NATIVE GUI EVALUATION STATUS

This appendix records why the method’s end-task claim is a boundary rather than a result. It is included because the manuscript should not be read as asserting an unmeasured gain.

**Controlled sealed test.** On a sealed template-C test of 200 prefixes, the adapted arm scores 23/200 against 24/200 for its matched control, with mean action NLL 0.412004 against 0.412617. Validation had selected the same arm at 0.18 against 0.16. This is a no-go for a discrete accuracy claim, and the sealed test was not reused for further selection.

**Native benchmarks.** The evaluated configuration is a frozen Qwen2.5-VL-7B-Instruct backbone with 930,305 trainable parameters (controller 856,577; residual 73,728), decoder-K intervention at layers 9–17, rank 8,  $\alpha = 8$ , and a single 256-wide GRU layer. Training used 200 controlled prefixes per arm with 50 validation prefixes; all six selected checkpoints reach 100% candidate accuracy on those validation prefixes, which is a controlled slot-selection number and not a task-success rate. At

the time of the log, completed episodes stood at 0/5 (new action protocol) and 0/19 (retired protocol) for AndroidWorld, and 0/21 for the adapted arm against 1/21 for cross-entropy on MemGUI under a registered alternate-judge protocol.

**Failure attribution.** These failures are not cleanly attributable to memory handling. In the AndroidWorld new-protocol episodes, all 150 executed steps were `answer` actions with zero illegal actions—the base policy never attempted a tap or text entry—so a first-step action error cannot be ascribed to history selection, and the first step has no historical image at all. Recorded infrastructure faults include an emulator clock regression of 34.5 hours after snapshot restore, a backend HTTP 500 on empty text input, and judge rate limiting; repair patches for the clock and gesture issues were verified but several landed mid-batch, so affected episodes use different effective environment revisions. Denominators here count completed episodes, not task types, and the batches are incomplete. We therefore report these numbers as a status record, not as a method comparison.

## F PRECURSOR RESULTS

These experiments concern earlier global-return and action-credit variants and are not mixed with SIGMA. They document why the final evaluation uses task success rather than confidence sharpening.

Table 5: Observed MiniWoB success counts. Each family has 100 common seeds, giving 400 paired task instances per method. All model differences occur in click-color; the other three families are unchanged or saturated.

| Task family     | Base  | Warm  | Global return | Success CE | Shuffle |
|-----------------|-------|-------|---------------|------------|---------|
| click-button    | 97    | 97    | 97            | 97         | 97      |
| button-sequence | 90    | 90    | 90            | 90         | 90      |
| click-color     | 42    | 43    | 43            | 46         | 41      |
| click-tab       | 100   | 100   | 100           | 100        | 100     |
| Total           | 329   | 330   | 330           | 333        | 328     |
| Success (%)     | 82.25 | 82.50 | 82.50         | 83.25      | 82.00   |

Table 5 gives the per-family counts. Global return versus success CE has zero rescues and three regressions, with exact paired McNemar  $p = .25$ ; versus the warm start it has two rescues and two regressions. The family-cluster interval versus CE is  $[-2.25, 0]$  percentage points based on four families and should not be read as a precise estimate for a broad population of GUI tasks. Mean candidate margins are .15017 for Base, .17110 for global return, and .14712 for CE.

Table 6: Observed AndroidWorld offline controls on six held-out trajectories. These are return-margin diagnostics, not task-success rates.

| Target              | Margin | Training range  | Margin |
|---------------------|--------|-----------------|--------|
| Warm start          | .13015 | $H = 1$         | .13975 |
| Normal return       | .14116 | $H = 3$         | .14387 |
| Shuffled return     | .13219 | $H = 5$         | .14304 |
| Random return       | .13745 | Full trajectory | .14309 |
| Sign-flipped return | .12312 | First-only      | .14663 |
|                     |        | Final-only      | .13843 |
|                     |        | Remove history  | .13786 |

Table 6 reports the returned-margin controls. Normal and sign-flipped labels move the offline score in opposite directions, but random labels also improve it and the first-only condition is strongest. These measurements motivate the stricter comparison of §5; they do not identify the cause of the precursor’s online outcome, and they do not prove that a correctly implemented on-policy return estimator cannot localise useful credit.

**Transport and gate observations.** An energy-regularised variant of the precursor reduces measured activation transport energy by  $4.47 \times (0.3004 \times 10^{-3} \rightarrow 0.0672 \times 10^{-3})$  while retaining 97.8% of

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the unregularised return margin. Its learned gate magnitudes over history slots are  $0.418 \rightarrow 0.179 \rightarrow 0.078$  for the distractor-credit family and  $0.704 \rightarrow 0.039 \rightarrow 0.290$  for the hidden-memory family: the first decays monotonically with distance from the current decision and the second does not. With 20–240 prefixes per cell and a single training run, we treat this as a descriptive observation about one trained controller, not as evidence for a general recency rule, and it is not used to support any main-text claim.

## G FIGURES AND DATA AVAILABILITY

Figures 1, 2, 3 and 5 are produced by matplotlib code in `plotting/`, with the shared visual system in `style.py`; the categorical and diverging palettes used there pass a colour-vision-deficiency separation check and every value is drawn from `data/`, which is generated by `plotting/extract_evidence.py`. The method diagram (Figure 4) is inline vector art in the source. No figure in this paper is generated imagery, and no missing value is rendered as a plotted point.

The supplementary archive contains the manuscript source, the plotting and extraction code, the extracted figure inputs, and the upstream experiment records the extraction reads. Running `extract_evidence.py` and then the four figure scripts regenerates every reported number and every quantitative figure from those records, and the extraction prints a verification report rather than silently defaulting a missing source. The archive contains no author-identifying information: paths are derived from `__file__`, and the staged tree is scanned for identifying strings before the archive is written.